Week 23 — Extended Reading: Shielding, barrier filters and robustness

STUDY-RL

Source audit: 20 October 2026

Read for a claim, an assumption and a test

Read the teaching guide first, then use this pack alongside the lecture. The annotations below are course-written paraphrases and investigation prompts, not quotations from the papers. Each source link identifies a primary paper or official guidance. Benchmark results do not transfer automatically to our small exercises. For each reading, write the mathematical object, the needed assumptions and one observation that would challenge your interpretation.

1 Alshiekh et al. (2018): Shielding

Primary source. https://arxiv.org/abs/1708.08611

Mechanism and scope. Shielding restricts decisions using a formal specification and an environment abstraction. Correctness depends on the abstraction and on remaining inside a region from which safe continuation is possible.

Read, derive, investigate. Draw a state that is safe now but has no safe successor action. Why does checking only the next label fail to establish indefinite safety without additional reachability assumptions? The course grid's first-admissible-action filter differs from sampling a renormalised masked policy, even though both can preserve the same invariant.

2 Dalal et al. (2018): Safety layers

Primary source. https://arxiv.org/abs/1801.08757

Mechanism and scope. A safety layer corrects proposed continuous actions using a learned local constraint model. The affine action projection is exact for the stated local inequality, not automatically for unknown nonlinear physical dynamics.

Read, derive, investigate. Derive the projection onto $b + w^\top a \leq 0$ with a Lagrange multiplier. Handle $w = 0$ explicitly: either every action is feasible or none is. Explain why adding epsilon to the denominator cannot create control authority or solve incompatible actuator bounds.

3 Tobin et al. (2017): Visual domain randomisation

Primary source. https://arxiv.org/abs/1703.06907

Mechanism and scope. This paper randomises simulated visual appearance for transfer of object localisation. It is useful historical context for domain randomisation, but it is not a proof about the pendulum's sampled mass and length dynamics.

Read, derive, investigate. Separate visual variation, physical parameter variation and worst-case robust optimisation. Averages under a chosen randomisation distribution do not bound unseen extremes. Propose a held-out parameter grid and report both average and worst observed return.

4 Sim-to-real survey (2025)

Primary source. https://arxiv.org/abs/2502.13187

Mechanism and scope. The survey maps sim-to-real approaches and potential roles for foundation models. A survey organises evidence from multiple settings; it does not establish that semantic representations universally remove the reality gap.

Read, derive, investigate. Choose one reviewed mechanism and identify its required observations, embodiment assumptions and evaluation domain. Which part could be represented in the course toy, and which would require a new experiment? Keep a claimed direction of research separate from a verified performance result.

5 Reed and Lahijanian (2024): Learned shielding

Primary source. https://arxiv.org/abs/2410.07359

Mechanism and scope. This work learns unknown dynamics with deep-kernel models and builds interval-MDP abstractions for shielding. Probabilistic assurance is tied to uncertainty estimation and formal synthesis assumptions.

Read, derive, investigate. Trace how a prediction interval becomes a transition-probability set. What happens if the interval loses coverage under distribution shift? The exact deterministic grid model bypasses this learning problem; its zero-violation demonstration does not reproduce unknown-dynamics assurances.

Connect the reading to your workbook

The notebook should distinguish execution safety from value estimation. The current tabular learner masks executed actions but maximises over all actions in its bootstrap. Compare that optimistic target with a maximum over safe next actions; preserve the terminal mask in both. An execution shield can still block a dangerous action while its unused optimistic value distorts learning.

Evidence boundary. Source identities and the central claims above were checked against primary material. These annotations do not reproduce every proof, training run or empirical result. Use the original paper for theorem assumptions and experimental details; use `project/` and the workbook for the exact course implementation.

Statistics reading bridge

Small per-decision model error can accumulate over a run. Read event probabilities and independence before composing per-decision claims. Separate a model-conditioned invariant, a

sampled failure rate and a horizon-level risk statement; use shielding primary sources for their precise model assumptions.

Checked textbook locations: Siegrist § § 2.4, 2.5, PDF pp.181, 197. https://stats.libretexts.org/@go/page/10133. These locations support the probability and estimation concepts; the algorithm-specific results remain attributed to the primary readings above. The course source catalog distinguishes the original textbook from its supplied instructor manual and records PDF-viewer pagination.